CS & IT ENGINEERING



Compiler Design

3AC, SSA, DAG

Code Optimisation

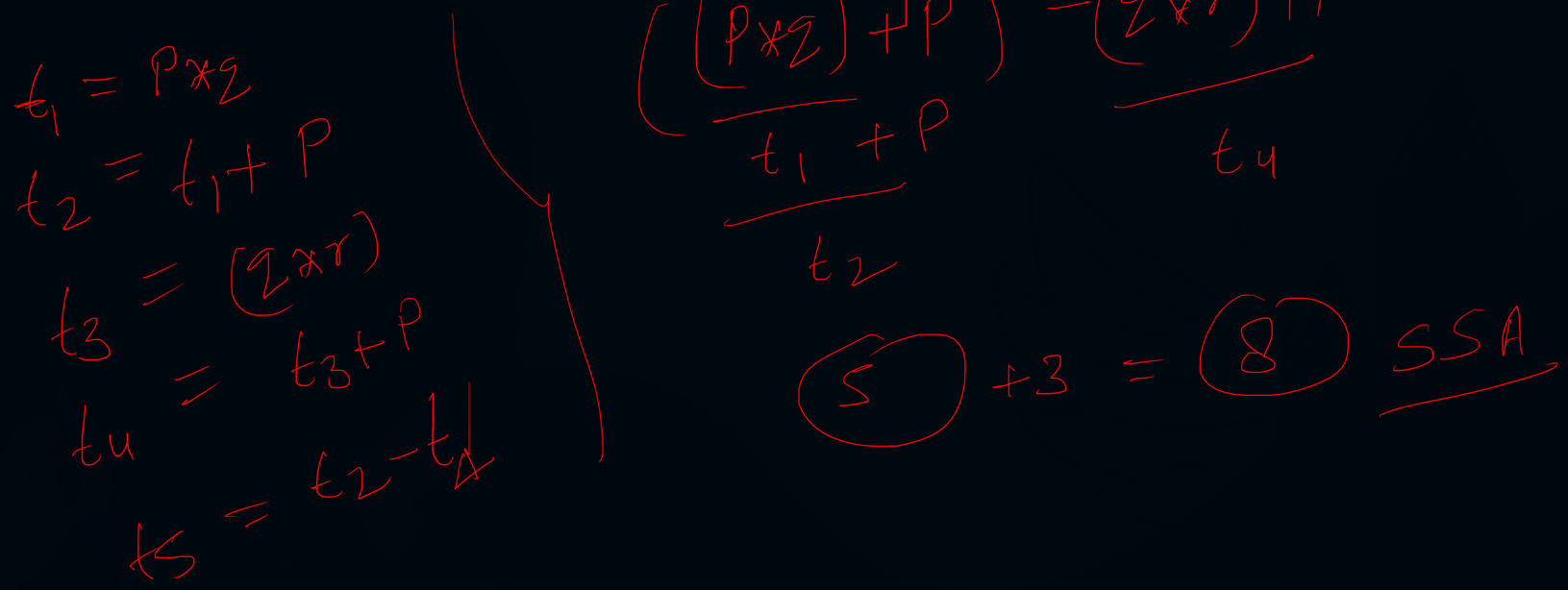
DPP – 01 Discussion Notes



By- Rohit Chauhan Sir

[NAT]

#Q. Consider the following expression: (((p * q) + p) – (q * r) + p)
The minimum number of variables required in the single static assignment code of above expression is ___8___.
[Note: Re - ordering is also possible.]



[NAT]

#Q. Consider the following expression ((a + b) * a) – ((b * c) + a).
Total minimum variables required for the above expression in SSA (static single assignment) form is/are ___.

$t_1 = a + b$
$t_2 = t_1 * a$
$t_3 = b * c$
$t_4 = t_3 + a$
$t_5 = t_2 - t_4$

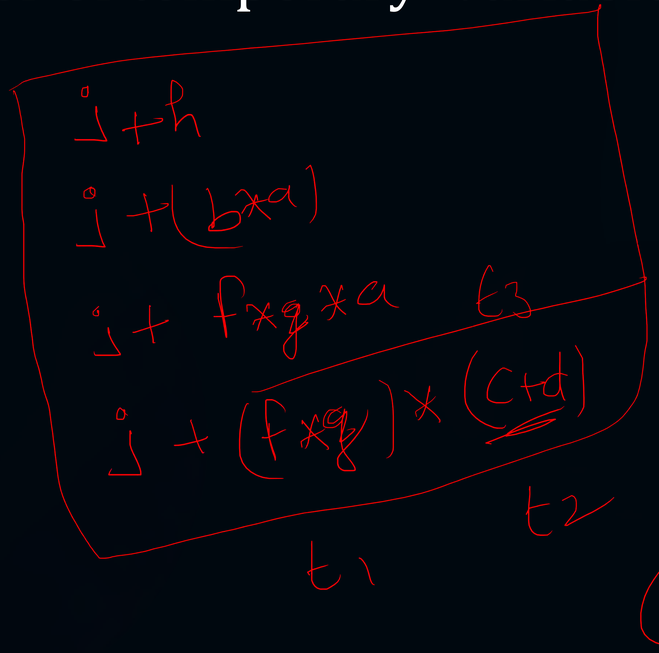
5 + 3 = 8

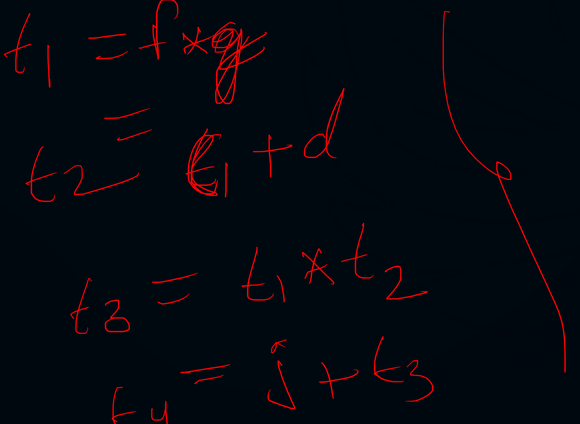
[NAT]

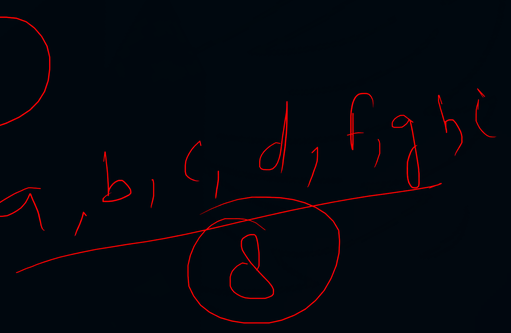
#Q. Consider the intermediate code given below:

a = c + d
b = f * g
h = b * a
i = j + h

If the above code is converted into static single assignment. The minimum number of temporary variables required is _____.







[MSQ]

#Q. Which of the following is true?

A. If CLR(1) does not contain SR conflict then LALR(1) don't have SR conflict.

B. If CLR(1) does not contain RR conflict then LALR(1) does not contain RR conflict.

C. If CLR (1) does not contain RR conflict then LALR (1) may contain RR conflict.

D. None

[MCQ]

#Q. Consider the following grammar:

$S \rightarrow a\ P\ Q\ R$

$P \rightarrow bc \mid \in$

$Q \rightarrow d \mid e \mid \in$

$R \rightarrow \in$

What the follow (P)?

$\{d, e, \$\}$

$Fo(P) = Fi(Q\ R)$

$= \{d, e, Fi(R)\}$

$= \{d, e, Fo(S)\}$

$= \{d, e, \$\}$

A {b, c, d, e}

B {b, c, d, e}

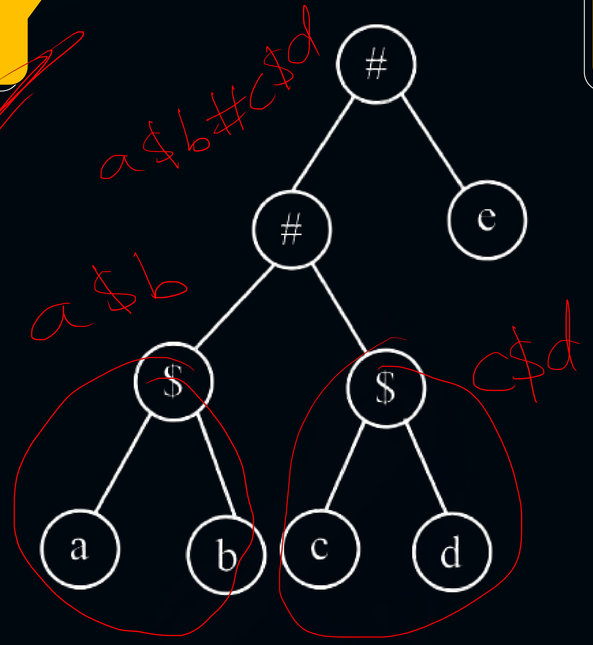
C {d, e, a, $}

D {d, e, $}

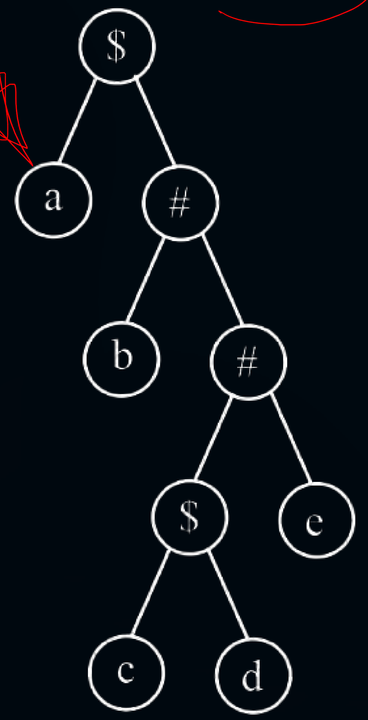
[MCQ]

#Q. Consider two binary operators $ and #. $ has higher precedence than #, $ is right associative while # is left associative. For expression is a $ b # c$ d # e. Which of the following representation of syntax tree for above expression?

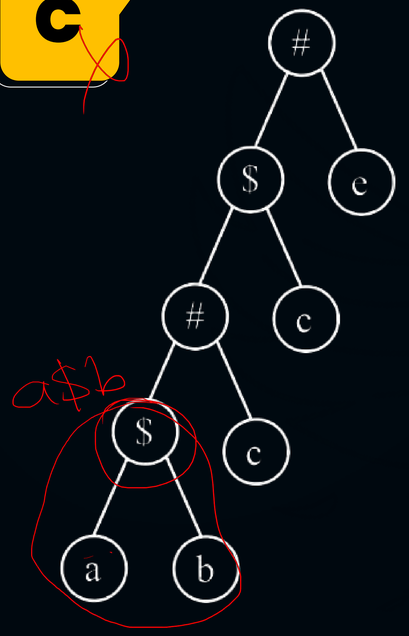
A



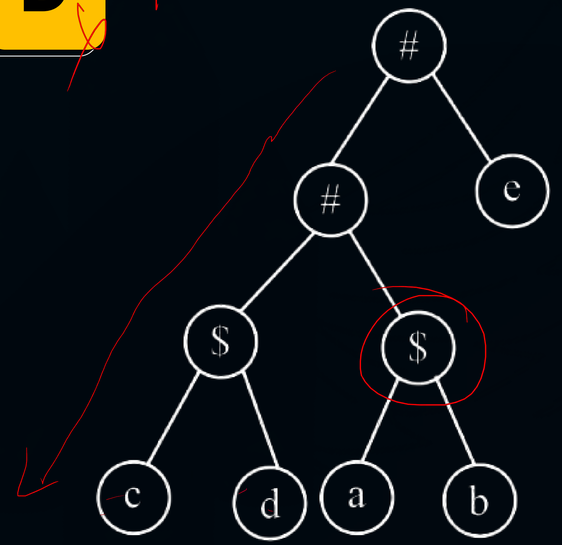
B



C

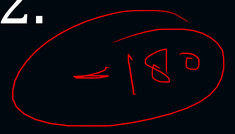


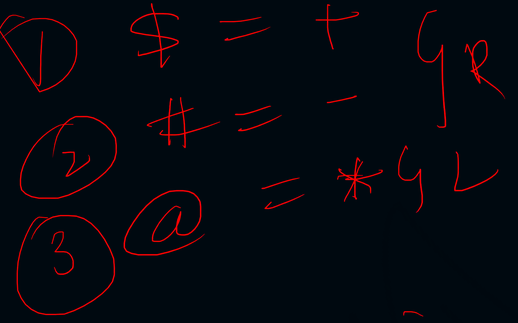
D



[NAT]

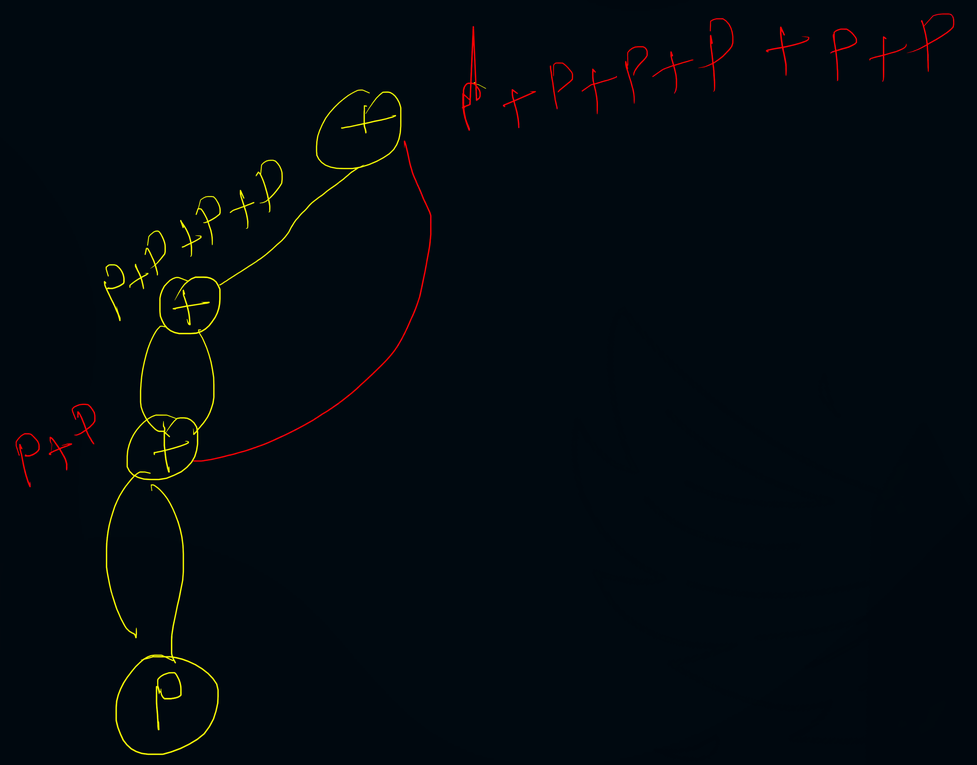
#Q. Consider 3 binary operators $, # and @, $ has higher precedence than # and # has higher precedence than @. The meaning of $, # and @ is +, –, and ×. # and $ are right associative whereas @ is left associative. The value of 6 # 7 # 1 $ 5 $ 3 @ 9 @ 2.





[MCQ]

#Q. Consider the following expression Q = P + P + P + P + P + P. How many nodes and edges are present in the direct acyclic graph (DAG) of intermediate code generation?



A. 8, 8

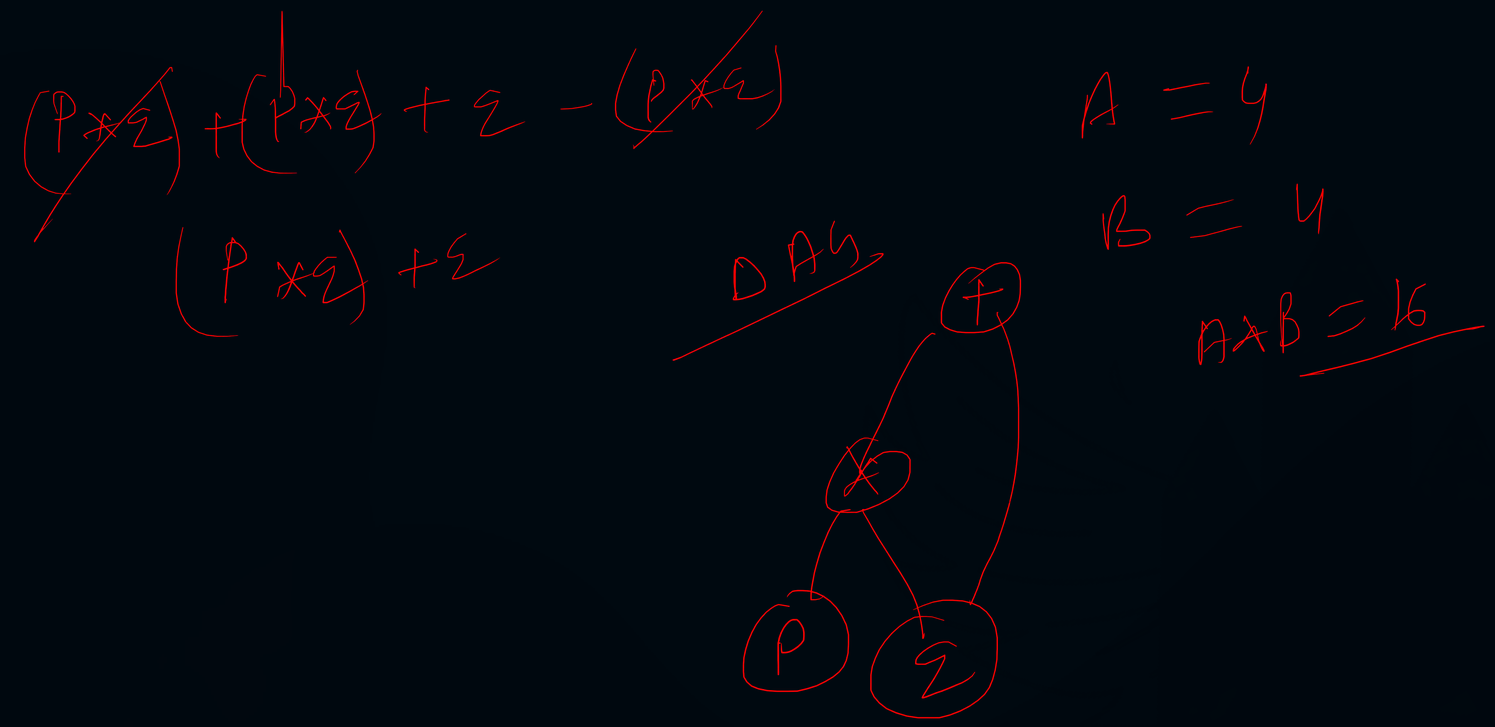
B. 4, 6

C. 10, 12

D. 8, 12

[NAT]

#Q. Consider the following expression: x = (p * q) + (p * q) + q – (p * q).
If total nodes in DAG (direct acyclic graph) is A and total edges in DAG is B then the value of A * B is

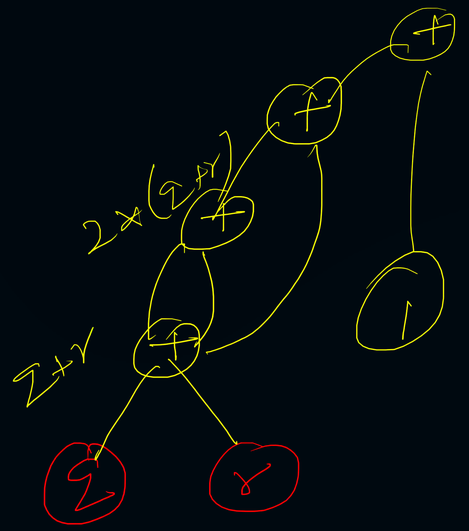


[MCQ]

#Q. Consider the following expression

p = q + r
s = p + 1
t = q + r
u = s + t
v = t + u

How many minimum number of nodes and edges are present in DAG?



A 7, 8

B 8, 7

C 9, 11

D 11, 9

(TO Problems

DAG, 3AC, SSA

ST

THANK - YOU